AI-guided programming concept mapping with AR for enhancing computational thinking

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Abstract: In the digital era, Computational Thinking (CT) is a core skill in programming education. However, students often struggle to understand its framework because the information is abstract and the logic is complex. Concept mapping is an effective tool to visualize this knowledge. However, the complex structure of concept maps can create an extra learning burden for students. To address this problem, this study developed a system that uses an Artificial Intelligence (AI) virtual learning assistant. This assistant guides students through the process of constructing concept maps for programming. The system uses Augmented Reality (AR) to provide an immersive experience. This design reduces the visual burden caused by switching between different devices. Ultimately, this study aims to improve student learning outcomes and interest through AI guidance. It helps students better understand the process of concept mapping and offers a new direction for programming instruction.

Keywords: Computational Thinking, Concept Mapping, AI Virtual Learning Assistant, AI-guided Concept Mapping

1. Introduction

Since 2018, the Ministry of Education in Taiwan has promoted innovation in programming education. It has also introduced courses to cultivate students' logical computational abilities. The goal is to improve students' CT and basic programming skills to meet the challenges of future technological development (National University of Kaohsiung Center for Instructional and Faculty Development, 2020).

Wing (2006) defined CT as a way to solve problems, design systems, and understand human behavior using fundamental computing concepts. Because CT relies on computing concepts, scholars believe that learning programming is an effective way to cultivate and realize CT skills (Pala & Türker, 2021; Shute et al., 2017).

However, learning programming is not an easy task. The structure of programming languages is complex. Many students feel frustrated when debugging errors. This frustration can reduce their interest in learning programming (Mannila et al., 2006).

To address these difficulties, some scholars suggest that incorporating appropriate guidance and feedback during the learning process helps students understand the programming flow and develop correct concepts. Therefore, research on concept mapping has received increasing attention in education. Novak (1984) proposed Concept Mapping as a learning strategy. It organizes fragmented information through an effective hierarchical arrangement. This process turns information into clear, visual learning materials (Clarke, 1991; Osman-Jouchoux, 1997).

Although concept mapping helps students organize and summarize knowledge, the process of absorbing and converting this information can still impose a learning burden. Therefore, a key issue in learning design is how to use concept mapping to assist learning while reducing students' cognitive load.

With the development of technology, many studies follow the Multimedia Learning Theory (Mayer, 2014). This theory establishes basic principles, such as the Spatial Contiguity Principle and the Navigation Principle. Within this framework, the application of AR offers a new support mechanism. First, regarding the Spatial Contiguity Principle, AR presents virtual text and images simultaneously. It overlays them onto the student's real environment. This combines virtual information with the real world. It reduces the visual burden of switching between different information sources and improves learning efficiency. Second, regarding the Navigation Principle, AR provides dynamic hints. These hints attract students' attention and guide them to focus on core learning content.

In addition, Lester et al. (1997) proposed the Persona Effect in multimedia presentation. They found that including a virtual character to assist teaching has a positive impact on students. Their research results show that students generally feel that such teaching assistants provide learning support and increase learning enjoyment.

Based on the above, this study combines an AI virtual learning assistant as a guiding tool. It uses the steps of CT to guide students. Specifically, it helps them understand programming logic and syntax while they construct concept maps. The assistant also provides immediate feedback and suggestions for correction. Through this integration of virtual and real elements, students can use the concept mapping mechanism to internalize the content into CT skills. This converts abstract ideas into concrete programming concepts and improves their understanding of programming logic and structure. Furthermore, the AI virtual learning assistant interacts with students in real-time. By answering questions or providing appropriate hints, it enhances engagement and enjoyment, thereby increasing students' commitment to learning programming.

Therefore, this study uses mobile devices to develop and operate the proposed learning system. It combines AR technology to present the AI virtual learning assistant in a 3D form within the physical environment. The study further investigates the differences in student performance in learning programming syntax under two conditions: with the AI virtual learning assistant and without it. Finally, this study evaluates the impact and benefits of the AI virtual learning assistant on programming learning.

2. Literature Review

2.1 Computational Thinking Guidance Mechanism Integrated with Concept Mapping

Since the development of concept maps, they have been regarded as a practical learning support tool. They help students and teachers organize and visualize knowledge structures. They also have diverse application values. These include deepening learning, designing teaching tools, assessing learning outcomes, clarifying misconceptions, and guiding teachers and students to explore the nature of knowledge construction (Novak, 1990). Past research in science education shows that concept mapping helps students deeply understand scientific concepts and achieve meaningful learning (Novak, Gowin & Johnson, 1983).

Based on the above, concept mapping helps students organize knowledge and understand the relationships between concepts. It can be applied to support and build students' CT skills. Brennan and Resnick (2012) studied the interactive programming tool Scratch. They defined CT concepts in programming and created a CT framework that includes concepts, practices, and perspectives. This organized thinking pattern parallels the core concepts of concept mapping. Concept mapping effectively presents CT in programming. It helps students master the steps of problem-solving. At the same time, it cultivates the ability to analyze and decompose problems. This further improves student learning outcomes.

2.2 AI Virtual Learning Assistant

An AI virtual learning assistant is a computer-assisted learning system. It monitors the user's learning progress and usage. It analyzes the problems encountered by the user and provides immediate assistance (Hsu, 2004). In addition, virtual assistants can be combined with other technologies based on different purposes, users, and themes. They are widely applied in various fields to help users obtain relevant information, improve work efficiency, and complete specific tasks (Zhan et al., 2014).

Cassell & Thórisson (1999) pointed out that introducing an AI virtual learning assistant into a learning system improves interaction and communication between students and the system. It also improves the fluency of information delivery. Pereira (2016) used an AI virtual learning assistant for conversational quizzes to help students learn autonomously. The results showed that 89% of students found this method effective. Furthermore, 72% of students stated that the virtual assistant helped them participate more actively in the course. They hoped for more interactive functions in the future to further enhance the learning experience.

Based on the literature review above, AI virtual learning assistants can improve interactivity and learning outcomes. Through immediate response and correction, they help students deeply understand and master knowledge structures. Therefore, this study combines an AI virtual learning assistant to guide students in filling out concept maps. During the process, students receive hints and learning content through the virtual character, rather than relying solely on text hints. At the same time, this study uses AR technology to present the AI virtual learning assistant and content. This creates an integrated virtual-real learning experience and further improves student engagement and interest.

3. Research Methodology

This study uses mobile devices as the primary hardware. The system is developed on the Android platform using Python. It integrates Firebase as the remote database to establish a comprehensive information service network.

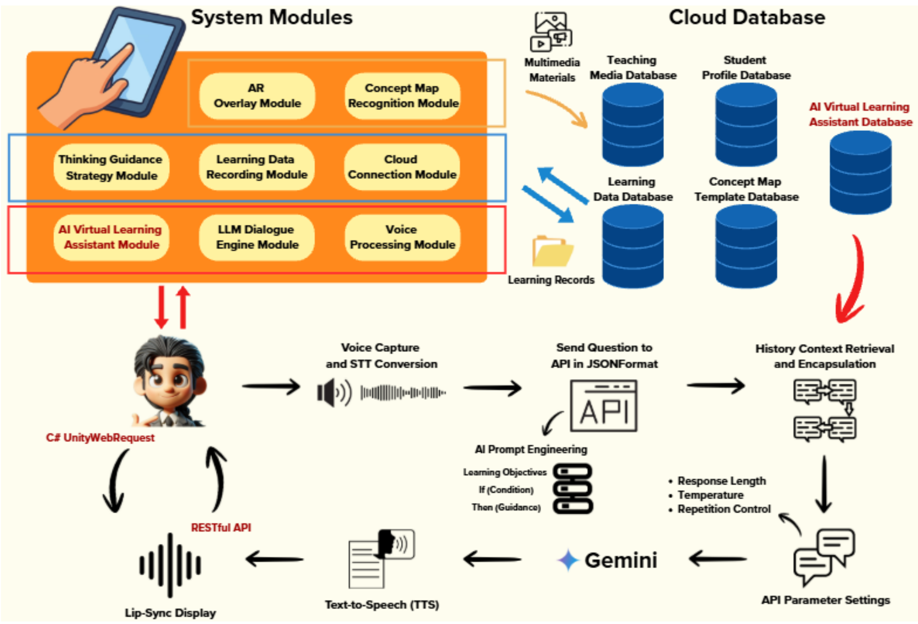

Figure 1. System Architecture

The objective of this study is to develop a system where an AI virtual learning assistant guides the construction of concept maps. The system architecture is illustrated in Figure 1.

The system utilizes AR technology to provide an immersive learning experience through virtual-real interaction. The AI virtual learning assistant guides students in filling out concept maps. Before the process begins, it provides clear directions to help students understand the structure. During the process, it offers hints. In addition, the system records the students' construction process and learning status. It collects data on the duration and frequency of interactions with the AI virtual learning assistant to analyze its impact on learning outcomes. Simultaneously, the system analyzes user habits through the learning record database to further optimize the learning system.

The system architecture consists of two main parts: AI Learning Assistant Guidance and Interaction and Database Management. The AI Learning Assistant Guidance and

Interaction part includes the Concept Map Recognition Module, Learning Data Recording Module, Cloud Connection Module, Thinking Guidance Strategy Module, AI Virtual Learning Assistant Module, LLM Dialogue Engine Module, Voice Processing Module and AR Presentation Module. The Database Management part includes the Concept Map Template Database, Learning Data Database, Student Profile Database, AI Virtual Learning Assistant Database, and Teaching Media Database.

3.1 System Modules

- Concept Map Recognition Module: Before the student begins filling out the concept map, this module uses markerless AR to scan the concept map topic on the paper via a mobile device. It then calls the Cloud Connection Module to retrieve the corresponding template and related learning materials from the Concept Map Template Database, as shown in Figure 2.

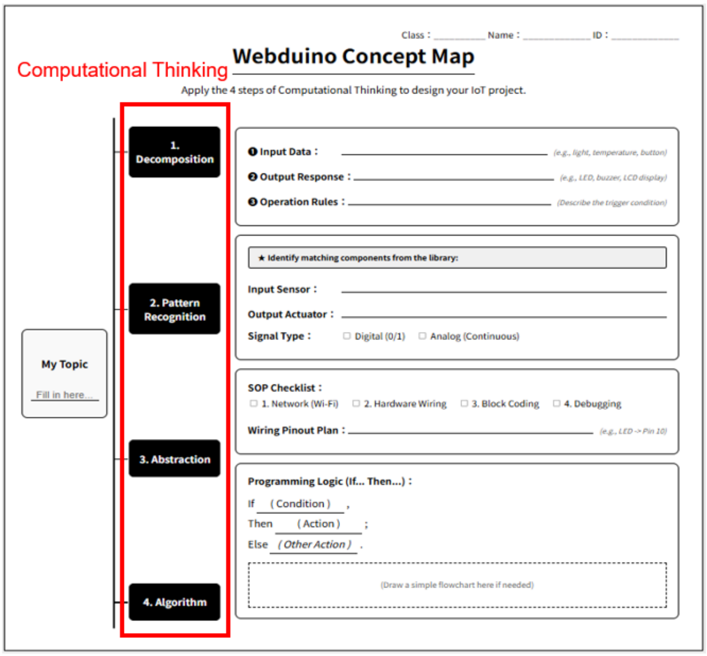

Figure 2. Example of a Webduino Concept Map Template

- Learning Data Recording Module: When a student fills out a concept map, this module records their individual learning process. This includes the time spent on each section and the frequency of interactions with the AI Virtual Learning Assistant. The collected data is stored in the Learning Data Database. This allows students to review their previous progress during their next session and enables the analysis of their learning history.
- Cloud Connection Module: This module connects to the cloud database to retrieve necessary teaching materials and templates. When a student scans the concept map template on paper, this module is triggered to obtain the corresponding template and teaching content. It then transmits these resources back to the system for the student to use.
- Thinking Guidance Strategy Module: When a student scans the concept map template on paper, this module uses the image generated by the Concept Map Recognition Module to project the AI Virtual Learning Assistant into the physical environment via AR. It then calls the Cloud Connection Module to retrieve corresponding teaching materials from the Teaching Media Database. The assistant guides the student through the filling process sequentially, following the four steps of CT. Furthermore, students can use a controller to select specific sections where they need guidance to help them complete the concept map, as shown in Figure 3.

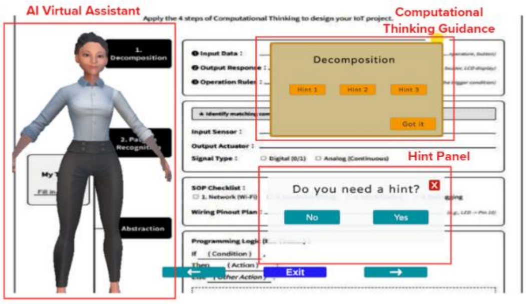

Figure 3. Thinking Guidance Interface

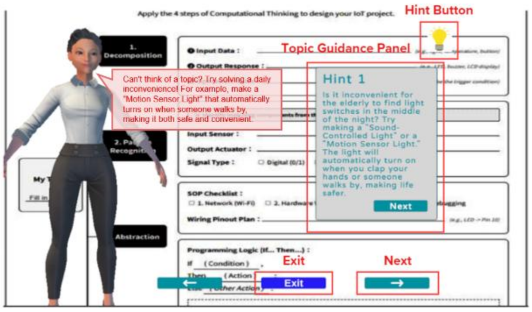

Figure 4. Dialogue Interface

- AI Virtual Learning Assistant Module: This module invokes the Cloud Connection Module to retrieve the AI Virtual Learning Assistant from the AI Virtual Learning Assistant Database for instructional guidance. Simultaneously, it connects to Gemini to engage in interactive dialogue with students. Once the input is transmitted to Gemini and a response is generated, it is sent back to the system. Furthermore, the module performs adaptive adjustments in subsequent responses to better align with the student's individualized instructional guidance strategies, as shown in Figure 4.
- LLM Dialogue Engine Module: Serving as the semantic processing core of the system, this module is responsible for receiving student input from the AI Virtual Learning Assistant Module and integrating it with preset thinking guidance prompts and historical dialogue context. This module invokes the Cloud Connection Module to transmit the consolidated information to the large language model, Gemini, for processing. Once the model generates a response, the module parses the unstructured reply into executable guidance commands for the system, ensuring that the content precisely corresponds to the four steps of CT.
- Voice Processing Module: This module aims to facilitate an intuitive spoken interaction experience by integrating Speech-to-Text (STT) and Text-to-Speech (TTS) technologies. When a student asks a question, this module invokes the Cloud Connection Module to convert audio data into text and transmits it to the LLM Dialogue Engine Module for semantic analysis. Conversely, once the system generates a response, this module converts the text into natural and fluid speech in real-time. This endows the virtual assistant within the scene with Q&A capabilities, thereby enhancing the sense of presence in the interaction.
- AR Presentation Module: This module projects the AI Virtual Learning Assistant into the physical environment. It uses Unity to design the 3D models and employs technologies such as skeletal animation and real-time lighting. These features ensure that the assistant's movements are smooth and adaptable to various environments. Regarding guidance, the module dynamically adjusts the assistant's actions and responses based on the student's learning process and interactions, thereby providing individualized learning support.

To specifically illustrate the role of the AI Virtual Learning Assistant within the system, a practical guidance scenario is described below. When a student struggles to define an IoT project topic during the "Decomposition" phase, traditional learning systems can only provide a static list of examples. However, the AI assistant in this system actively intervenes. Through the LLM Dialogue Engine, the AI assistant analyzes the step where the student is currently stalled and generates contextual guidance prompts, such as: "Is it inconvenient for the elderly to find switches in the middle of the night? Try making a sound-controlled light!"

If the student is confused by this prompt and asks via voice input, "But I don't know what sensor a sound-controlled light needs," the AI assistant does not provide a direct answer. Instead, it responds using Socratic questioning: "Think about it, what kind of electronic component do we need to receive sound? Just like human ears!" Through this real-time, unscripted natural language dialogue, the AI assistant plays the role of a personalized tutor providing instructional scaffolding.

4. Conclusion and Future Work

This study has successfully developed a programming concept mapping system integrating an AI Virtual Learning Assistant with AR technology. The system features step-by-step thinking guidance, real-time interactive dialogue, and the recording of learning portfolios. Currently, system development is complete, and the next phase will validate its instructional effectiveness through a quasi-experimental design.

Subsequent experiments will compare the differences between the experimental group (receiving AI assistant guidance) and the control group (receiving only general thinking guidance). Through pre-tests, post-tests, and relevant questionnaires, this research will comprehensively evaluate the impact of this learning mode on several key variables: students' learning achievements, learning motivation, CT skills, cognitive load, and system satisfaction. Ultimately, this study anticipates that the system will effectively assist students in internalizing CT and practically applying it to solve programming problems.

References

Brennan, K., & Resnick, M. (2012, April). *New frameworks for studying and assessing the development of computational thinking* [Paper presentation]. 2012 Annual Meeting of the American Educational Research Association, Vancouver, BC, Canada.

Cassell, J., & Thórisson, K. (1999). The power of a nod and a glance: Envelope vs. emotional. *Applied Artificial Intelligence, 13*, 519–538.

Center for Instructional Development, National University of Kaohsiung. (2020). *Explanation of recognition standards for the Ministry of Education's promotion of "Programming and Computational Thinking Related Courses" in universities*.

Clarke, J. H. (1991). Using visual organizers to focus on thinking. *Journal of Reading, 34*(7), 526–534. Conger, S., & Loch, K. D. (1995). Ethics and computer use. *Communications of the ACM*, *38*(12), 30-32.

Hsu, C.-N. (2004, October 30). Intelligent agents, omnipotent? *CTIMES*. https://www.ctimes.com.tw/DispArt/tw/%E7%B6%B2%E9%9A%9B%E7%AE%A1%E7%90%86%E7%B3%BB%E7%B5%B1/0410301612GK.shtml

Lester, J. C., Converse, S. A., Kahler, S. E., Barlow, S. T., Stone, B. A., & Bhogal, R. S. (1997). The persona effect: Affective impact of animated pedagogical agents. *Proceedings of the ACM SIGCHI Conference on Human Factors in Computing Systems*, 359–366.

Mannila, L., Peltomäki, M., & Salakoski, T. (2006). What about a simple language? Analyzing the difficulties in learning to program. *Computer Science Education, 16*(3), 211–227.

Mayer, R. E. (2014). Cognitive theory of multimedia learning. In R. E. Mayer (Ed.), *The Cambridge handbook of multimedia learning* (pp. 43–71). Cambridge University Press. https://doi.org/10.1017/CBO9781139547369.005

Novak, J. D. (1990). Concept mapping: A useful tool for science education. *Journal of Research in Science Teaching, 27*(10), 937–949.

Novak, J. D., & Gowin, D. B. (1984). *Learning how to learn*. Cambridge University Press.

Novak, J. D., Gowin, D. B., & Johnson, G. T. (1983). The use of concept mapping and knowledge Vee mapping with junior high school science students. *Science Education, 67*(5), 625–645.

Osman-Jouchoux, R. (1997). *Linking reading and writing: Concept mapping as an organizing tactic* (ERIC Document Reproduction Service No. ED408955).

Pala, F. K., & Türker, P. M. (2021). The effects of different programming trainings on the computational thinking skills. *Interactive Learning Environments, 29*(7), 1090–1100. https://doi.org/10.1080/10494820.2019.1635495

Pereira, J. (2016). Leveraging chatbots to improve self-guided learning through conversational quizzes. *Proceedings of the 4th International Conference on Technological Ecosystems for Enhancing Multiculturality (TEEM)*, 911–918.

Shute, V. J., Sun, C., & Asbell-Clarke, J. (2017). Demystifying computational thinking. *Educational Research Review, 22*, 142–158. https://doi.org/10.1016/j.edurev.2017.09.003

Wing, J. M. (2006). Computational thinking. *Communications of the ACM, 49*(3), 33–35.

Zhan, Z.-H., Liang, T., & Ma, Z.-C. (2014). Virtual assistant-based distance learning support services and technical difficulties. *Modern Distance Education Research, 6*, 95–103.